

Complete One-Sample T-Test — Steps 1 to 9

Example: A population has an average IQ of 100. Researchers want to determine whether a new medication affects intelligence.

Given	Value
Population / hypothesized mean (μ)	100
Sample size (n)	30
Sample mean ($\bar{x}$)	140
Sample standard deviation (s)	20
Confidence level	95%
Significance level (α)	0.05

Step 1: State the Hypotheses

- **Null Hypothesis (H_0):** The medication has no effect on average IQ.

$$H_0: \mu = 100$$

- **Alternative Hypothesis (H_a):** The medication affects average IQ.

- $H_a: \mu \neq 100$

- Because the alternative hypothesis uses $\neq$, this is a **two-tailed test**.

Step 2: Identify the Test

- A **one-sample T-test** is appropriate because we are comparing one sample mean with a hypothesized population mean, and the population standard deviation is unknown.

Use the sample standard deviation $s = 20$.

Step 3: Calculate Degrees of Freedom

- For a one-sample T-test:

$$df = n - 1$$

- $df = 30 - 1 = 29$

Degrees of freedom = 29

Step 4: Find the Critical T-Value

- Significance level: $\alpha = 0.05$

- Because this is a two-tailed test, divide α equally between the two tails:

$$\alpha/2 = 0.05/2 = 0.025$$

- Using a T-table with **df = 29** and a **two-tailed $\alpha = 0.05$** :

$$t\text{-critical} = \pm 2.045$$

- Rejection regions: $t < -2.045$ or $t > +2.045$.

Step 5: Calculate the Standard Error

- Formula:

$$SE = s / \sqrt{n}$$

- Substitute the values:

$$SE = 20 / \sqrt{30}$$

- $\sqrt{30} \approx 5.477$

$$SE \approx 20 / 5.477 = 3.651$$

Step 6: Calculate the T-Statistic

- Formula:

$$t = (\bar{x} - \mu) / (s / \sqrt{n})$$

- Substitute:

$$t = (140 - 100) / (20 / \sqrt{30})$$

- Numerator: $140 - 100 = 40$

- Standard error ≈ 3.651

$$t = 40 / 3.651 \approx 10.954$$

- **Calculated $t \approx 10.954$**

Step 7: Compare Calculated T with Critical T

- Calculated $t = 10.954$

- Critical $t = \pm 2.045$

- Since:

- **$10.954 > 2.045$**

- The calculated t-value lies in the rejection region.

- **Decision: Reject H_0 .**

Step 8: Calculate the P-Value

- For $t \approx 10.954$ with $df = 29$ in a two-tailed test, the p-value is approximately:

- **$p \approx 8.02 \times 10^{-12}$**

- This is far smaller than $\alpha = 0.05$:

- **$p < 0.05$**

- **Decision: Reject H_0 .**

Step 9: Final Conclusion

- At the 5% significance level, there is sufficient statistical evidence to conclude that the medication has a statistically significant effect on average IQ.

- Because the sample mean is 140, which is greater than the hypothesized mean of 100, the observed effect is an **increase** in IQ.

- **Final conclusion:** Reject H_0 . The sample provides strong evidence that the population mean IQ after taking the medication is different from 100.

Quick T-Test Flow: Hypotheses → Two-tailed test → $df = n-1$ → choose α → T-table → critical t → standard error → calculated t → compare → decision.

Important: The critical t-value is determined by the degrees of freedom, significance level, and whether the test is one-tailed or two-tailed. It is not selected from the calculated t-value.